Macroeconomics I — MPE Insper

Practice list: General Equilibrium, Money and Inflation — graduate level (Listas 5–6)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — General equilibrium and the neoclassical growth model

(Kurlat ch. 9; Romer ch. 2 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A two-period economy has a fixed capital stock K and no investment. Households have $u(c) + \psi \ln l$ each period, $u = c^{1-\sigma}/(1-\sigma)$, one unit of time ($l + L = 1$); firms produce $Y = AK^{\alpha}L^{1-\alpha}$ in competitive markets. In equilibrium, with $L = 1/3$, $\sigma = 2$ and $\alpha = 1/3$, the elasticity $d\ln L/d\ln A$ is:

A. -0.46

B. 0

C. $+0.46$

D. -1.00

Answer: A

Step — equilibrium conditions. Goods clearing without investment: $c = Y$. Firm: $w = (1-\alpha)Y/L$. Household: $\psi/(1-L) = w\,u'(c)$. Combining: $\dfrac{\psi}{1-L} = \dfrac{(1-\alpha)Y^{1-\sigma}}{L}$.

Step — log-differentiation. $\frac{L}{1-L}d\ln L = (1-\sigma)d\ln Y - d\ln L$ and $d\ln Y = d\ln A + (1-\alpha)d\ln L$:

$$\frac{d\ln L}{d\ln A} = \frac{1-\sigma}{L/(1-L) + \alpha + (1-\alpha)\sigma} = \frac{-1}{0.5 + 1/3 + 4/3} = -0.46.$$

With $\sigma > 1$ the income effect dominates; B is the log case.

Ref: Kurlat (2020), §9.1, pp. 165–168; Lista 5, Q1(e).

b) (3 points) In the continuous-time Ramsey model with $f(\tilde{k}) = \tilde{k}^{1/3}$, $\rho = 0.03$, $\sigma = 2$, $g = 0.02$, $n = 0.01$, $\delta = 0.05$, the steady-state gross saving rate $(n+g+\delta)\tilde{k}^*/f(\tilde{k}^*)$ is:

A. 33.3%

B. 22.2%

C. 26.7%

D. 16.7%

Answer: B

Step — modified golden rule. On the balanced path $\dot{\tilde{c}} = 0$: $f'(\tilde{k}^*) = \rho + \delta + \sigma g = 0.12$.

Step — saving rate. $s = \dfrac{(n+g+\delta)\tilde{k}}{f(\tilde{k})} = \dfrac{(n+g+\delta)\alpha}{f'(\tilde{k})} = \dfrac{0.08/3}{0.12} = 22.2\%$, below the Golden Rule $\alpha = 33.3\%$ (A). C uses $\sigma = 1$.

Ref: Romer (2012), §2.5, and problem 2.6; Kurlat (2020), §9.3, p. 177.

c) (2 points, T or F) In the Ramsey model, a permanent, unanticipated rise in government purchases financed by lump-sum taxes lowers consumption one for one on impact and leaves the capital stock unchanged forever.

Answer: T

Step — why. $\dot{\tilde{k}} = f(\tilde{k}) - \hat{c} - \hat{G} - (n+g+\delta)\tilde{k}$; the Euler equation does not involve $\hat{G}$. The new steady state has the same $\tilde{k}^*$ and $\hat{c}^*$ lower by $\Delta\hat{G}$; the new saddle path is the old one shifted down by $\Delta\hat{G}$, so $\hat{c}$ jumps by $-\Delta\hat{G}$ and there is no transition.

Ref: Romer (2012), §2.7, pp. 71–76.

d) (2 points, T or F) A permanent tax on capital income whose revenue is rebated lump-sum leaves the steady-state capital stock unchanged, because the rebate gives the revenue back.
Answer: F

Step — wedge. The rebate removes the income effect, not the wedge: the Euler equation uses the after-tax return, $(1-\tau)(f'-\delta) = \rho + \sigma g$, so $f'(\tilde{k}^*)$ rises and $\tilde{k}^*$ falls (Question 2(d)).

Ref: Romer (2012), problem 2.10; Kurlat (2020), Exercise 9.6.

Question 2. (15 points) Discursive. The Ramsey–Cass–Koopmans model. Households (population growing at n) maximize $\int_0^\infty e^{-\rho t}u(c)\,dt$, $u = c^{1-\sigma}/(1-\sigma)$, own the capital and rent it to firms; firms produce $Y = F(K, AL)$ with constant returns, A grows at g; capital depreciates at δ. Assume $\rho > n + (1-\sigma)g$. Notation: $\tilde{k} = K/(AL)$, $\hat{c} = c/A$, $f(\tilde{k}) = F(\tilde{k}, 1)$. Numbers: $f = \tilde{k}^{1/3}$, $\rho = 0.03$, $\sigma = 2$, $g = 0.02$, $n = 0.01$, $\delta = 0.05$.

a) (3 points) Derive the Euler equation from the discrete-time condition $u'(c_t) = e^{-\rho\Delta}(1 + r_{t+\Delta}\Delta)u'(c_{t+\Delta})$ (periods of length Δ) by letting $\Delta \to 0$. Use the firms' problem and no-arbitrage to get $r = f'(\tilde{k}) - \delta$, and write the system in $(\tilde{k}, \hat{c})$.

Step — limit. With CRRA, $c_t^{-\sigma} = e^{-\rho\Delta}(1 + r\Delta)c_{t+\Delta}^{-\sigma}$. Taking logs: $\sigma[\ln c_{t+\Delta} - \ln c_t] = \ln(1 + r\Delta) - \rho\Delta$. Divide by Δ and let $\Delta \to 0$:

$$\sigma\frac{\dot{c}}{c} = r - \rho \quad \Longrightarrow \quad \frac{\dot{c}}{c} = \frac{r-\rho}{\sigma}.$$

Step — firms and no-arbitrage. Profit maximization: $r^K = f'(\tilde{k})$, $w = A[f(\tilde{k}) - \tilde{k}f'(\tilde{k})]$. Holding capital pays $r^K - \delta$, holding bonds pays r; both are held only if $r = f'(\tilde{k}) - \delta$.

Step — system. With $\hat{c} = c/A$, $\dot{\hat{c}}/\hat{c} = \dot{c}/c - g$. Goods clearing, $\dot{K} = Y - C - \delta K$, in effective units:

$$\dot{\tilde{k}} = f(\tilde{k}) - \hat{c} - (n+g+\delta)\tilde{k}, \qquad \frac{\dot{\hat{c}}}{\hat{c}} = \frac{f'(\tilde{k}) - \delta - \rho - \sigma g}{\sigma}.$$

A planner maximizing the same utility subject to the same resource constraint gets the same two equations: the equilibrium is efficient (First Welfare Theorem).

Ref: Romer (2012), §2.2–2.4; Kurlat (2020), §9.3, eqs. (9.3.14)–(9.3.15).

Grading rubric.

- Limit of the discrete Euler equation, with the log step: 1 pt. Quoting $\dot{c}/c = (r-\rho)/\sigma$: max. 0.5 pt here.
- Firms and no-arbitrage: 1 pt.
- System in effective units: 1 pt.

b) (4 points) Prove that the interior steady state is unique and that $\tilde{k}^*$ lies below the Golden Rule. Compute $\tilde{k}^*$, $\hat{c}^*$, r^*, the saving rate and $\tilde{k}_{gold}$.

Step — uniqueness. $\dot{\hat{c}} = 0$ with $\hat{c} > 0$ requires $f'(\tilde{k}^*) = \rho + \delta + \sigma g$. Since f' is continuous and strictly decreasing from ∞ to 0 (Inada), there is exactly one such $\tilde{k}^*$; then $\hat{c}^* = f(\tilde{k}^*) - (n+g+\delta)\tilde{k}^*$.

Step — below the Golden Rule. The Golden Rule has $f'(\tilde{k}_{gold}) = n + g + \delta$. The assumption $\rho > n + (1-\sigma)g$ is $\rho + \sigma g > n + g$, so $f'(\tilde{k}^*) > f'(\tilde{k}_{gold})$ and, f' being decreasing, $\tilde{k}^* < \tilde{k}_{gold}$.

Step — numbers. $f'(\tilde{k}^*) = 0.12$: $\tilde{k}^* = (\frac{1/3}{0.12})^{3/2} = 4.630$, $f = 1.667$, $\hat{c}^* = 1.667 - 0.08 \cdot 4.630 = 1.296$, $r^* = 0.07$, $s = 0.370/1.667 = 22.2\%$; $\tilde{k}_{gold} = (\frac{1/3}{0.08})^{3/2} = 8.505$.

Economic intuition. Impatience has value: building capital up to the Golden Rule would sacrifice consumption now for more later, and households discount the later gain at $\rho + \sigma g$.

Ref: Romer (2012), §2.5; Kurlat (2020), §9.3, p. 177.

Grading rubric.

- Uniqueness argument: 1 pt.
- Proof that $\tilde{k}^* < \tilde{k}_{gold}$ using the assumption: 1.5 pts. Comparison by numbers only: 0.5 pt.
- Numbers: 1.5 pts.

c) (4 points) Linearize the system around the steady state. Show that it is a saddle, give the eigenvalues, the slope of the stable arm and the half-life of the gap. Compare with a Solow economy saving the same rate. Then set $g = 0$ and prove that the speed of convergence falls with σ, with limits as $\sigma \to 0$ and $\sigma \to \infty$.

Step — Jacobian. At the steady state, with $b \equiv f'(\tilde{k}^*) - (n + g + \delta) = \rho + \sigma g - n - g > 0$:

$$J = \begin{pmatrix} b & -1 \\ \hat{c}^* f''(\tilde{k}^*)/\sigma & 0 \end{pmatrix}, \qquad \lambda^2 - b\lambda + \hat{c}^* f''(\tilde{k}^*)/\sigma = 0.$$

Step — saddle. $\lambda_1 \lambda_2 = \hat{c}^* f''/\sigma < 0$: real roots of opposite sign. One predetermined variable ($\tilde{k}$) and one jump variable ($\hat{c}$) give a unique convergent path.

Step — numbers. $b = 0.04$, $f'' = \alpha(\alpha - 1)\tilde{k}^{*\alpha-2} = -0.01728$, $\hat{c}^* f''/\sigma = -0.0112$: $\lambda_\pm = \frac{1}{2}[0.04 \pm \sqrt{0.0016 + 0.0448}]$, so $\lambda_- = -0.0877$, $\lambda_+ = 0.1277$.

Step — stable arm. From the first row, $\lambda_-(\tilde{k} - \tilde{k}^*) = b(\tilde{k} - \tilde{k}^*) - (\hat{c} - \hat{c}^*)$, so $\hat{c} - \hat{c}^* = (b - \lambda_-)(\tilde{k} - \tilde{k}^*)$, slope $b - \lambda_- = \lambda_+ = 0.128$. Half-life $\ln 2/0.0877 = 7.9$ years. A Solow economy with $s = 22.2\%$ converges at $(1 - \alpha)(n + g + \delta) = 5.3\%$: Ramsey households save more when $\tilde{k}$ is low (r high), which speeds convergence.

Step — $g = 0$. Then $\tilde{k}^*$ ($f' = \rho + \delta$), $\hat{c}^*$ and $b = \rho - n$ do not depend on σ, and $|\lambda_-| = \frac{1}{2}\left[\sqrt{b^2 + 4\hat{c}^*|f''|/\sigma} - b\right]$ is strictly decreasing in σ, with $|\lambda_-| \to \infty$ as $\sigma \to 0$ and $|\lambda_-| \to 0$ as $\sigma \to \infty$: a household unwilling to tilt consumption never accumulates.

Ref: Romer (2012), §2.6, eqs. (2.26)–(2.39), pp. 67–71, and problem 2.8.

Grading rubric.

- Jacobian and characteristic polynomial: 1 pt.
- Saddle argument (sign of the determinant, jump variable): 1 pt. "It is a saddle" with no argument: 0 for this line.
- Eigenvalues, slope and half-life: 1 pt.
- $g = 0$ monotonicity and limits: 1 pt.

d) (4 points) At $t = 0$ the government unexpectedly and permanently taxes the net return to capital at $\tau = 0.25$, rebating the revenue lump-sum. Find the new $\tilde{k}^*$ and $\hat{c}^*$. Use the linearized stable arm at the new steady state to approximate the jump in $\hat{c}(0)$; the exact value (numerical saddle path) is $\hat{c}(0) = 1.391$. Describe the transition, and say what changes if the tax is announced at $t = 0$ to start at $T > 0$.

Step — new steady state. Households earn $(1 - \tau)(f' - \delta)$: $f'(\tilde{k}^*) = \delta + \frac{\rho + \sigma g}{1 - \tau} = 0.1433$, $\tilde{k}^* = 3.546$ (-23.4%), $\hat{c}^* = 1.241$ (-4.2%). The rebate keeps the resource constraint unchanged.

Step — linear jump. At the new steady state $b' = 0.1433 - 0.08 = 0.0633$; the Euler row now carries $(1 - \tau)$: $\lambda^2 - b'\lambda + (1 - \tau)\hat{c}^* f''/\sigma = 0$, $\lambda_- = -0.0847$, slope $b' - \lambda_- = 0.148$. $\hat{c}(0) \approx 1.241 + 0.148(4.630 - 3.546) = 1.402$, i.e. $+8.1\%$ above the old $\hat{c}^*$.

Step — exact vs linear. The exact jump is to 1.391 ($+7.3\%$). The 0.8% error comes from the size of the gap ($\tilde{k}$ starts 31% above the new steady state), where curvature matters.

Step — transition. $\hat{c}$ jumps up (the return to saving falls), then $\tilde{k}$ and $\hat{c}$ fall together along the stable arm to a lower steady state: consumption is front-loaded.

Step — announced tax. Consumption can jump only when news arrives. At $t = 0$ $\hat{c}$ jumps; on $[0, T)$ the economy follows the *old* dynamics on an unstable path chosen to reach the new stable arm exactly at T; no jump occurs at T.

Ref: Romer (2012), problems 2.10–2.12, pp. 95–96; Kurlat (2020), Exercise 9.6.

Grading rubric.

- New steady state: 1 pt. Using $f' = \rho + \delta + \sigma g$ (no tax): 0 for this line.
- Linearized jump with the $(1 - \tau)$ factor: 1.5 pts.
- Transition: 0.5 pt.
- Announcement: 1 pt. A jump at T: 0 for this line.

Block 2 — Money, inflation and seigniorage

(Kurlat chs. 10–11; Romer ch. 11 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) Baumol–Tobin: income $Y = 1200$ a year, cost $F = 2$ per trip to the bank, nominal rate $i = 6\%$. The welfare cost of $i > 0$, measured as the Bailey area under the money-demand curve $(\int_0^i m(x)\,dx - i\,m(i))$, is:

A. 8.49

B. 16.97

C. 4.24

D. 0

Answer: A

Step — demand. Minimize $FN + iY/(2N)$: $N^* = \sqrt{iY/(2F)} = 4.243$, $m = Y/(2N^*) = \sqrt{FY/(2i)} = 141.4$.

Step — Bailey area. $\int_0^i \sqrt{FY/2}\,x^{-1/2}dx - i\,m(i) = 2\sqrt{FYi/2} - \sqrt{FYi/2} = \sqrt{FYi/2} = 8.49 = FN^*$: exactly the resources spent on trips. The forgone interest $i\,m = 8.49$ is a transfer to the issuer, not a loss; B adds it.

Ref: Kurlat (2020), §10.4, eqs. (10.4.1)–(10.4.4), pp. 199–201; §11.4, pp. 217–218.

b) (3 points) Cagan with perfect foresight, in logs: $m_t - p_t = -a(p_{t+1} - p_t)$, $a = 4$ (monthly, as in Kurlat's Exercise 11.6). At $t = 0$ the central bank credibly announces a permanent 10% rise in the money supply at $t = 4$. On impact, p_0 rises by:

A. 0%

B. 4.1%

C. 10%

D. 8.0%

Answer: B

Step — forward solution. $p_t = \frac{m_t}{1+a} + \frac{a}{1+a}p_{t+1}$; iterating forward with no bubbles $(\lim(\frac{a}{1+a})^j p_{t+j} = 0)$: $p_t = \frac{1}{1+a}\sum_{j\geq 0}(\frac{a}{1+a})^j m_{t+j}$.

Step — impact. $\Delta p_0 = \frac{1}{1+a}\sum_{j\geq 4}(\frac{a}{1+a})^j \cdot 10\% = (\frac{a}{1+a})^4 \cdot 10\% = 0.8^4 \cdot 10\% = 4.1\%$.

Ref: Kurlat (2020), Exercise 11.6, pp. 219–221; Romer (2012), problem 11.1, p. 576.

c) (2 points, T or F) In the Cagan model with perfect foresight, a credible announcement that money growth will fall permanently in the future leaves inflation unchanged until money growth actually falls.

Answer: F

Step — forward-looking. By the forward solution, a lower future money path lowers p_t today; expected inflation falls, real money demand rises and inflation falls on announcement.

Ref: Romer (2012), problem 11.1; Kurlat (2020), Exercise 11.6 (stabilization).

d) (2 points, T or F) If nominal interest income is taxed at 30%, keeping the after-tax real return constant requires the nominal rate to rise by more than one point per point of expected inflation.

Answer: T

Step — Darby effect. $(1-t)i - \pi^e = \text{const} \Rightarrow di/d\pi^e = 1/(1-t) = 1.43$.

Ref: Kurlat (2020), §11.4, pp. 217–218.

Question 4. (15 points) Discursive. Seigniorage and hyperinflation, Brazil in the 1980s (illustrative). Real money demand relative to GDP is $m = \kappa e^{-a\pi^e}$ (κ absorbs the real rate); output is constant; a deficit d (share of GDP) is financed by printing money, $\mu m = d$ with $\mu = \dot{M}/M$; money market clearing gives $\dot{m}/m = \mu - \pi$. Numbers: $\kappa = 0.3$, $a = 2.5$, $d = 0.03$.

a) (3 points) Characterize the steady states. Find the revenue-maximizing inflation and the maximum deficit d_{max}, prove that there are exactly two steady states when $0 < d < d_{max}$, and compute them.

Step — steady state. $\dot{m} = 0$ gives $\pi = \mu$ and $\pi^e = \pi$, so $d = S(\pi) \equiv \pi\kappa e^{-a\pi}$.

Step — Laffer curve. $S'(\pi) = \kappa e^{-a\pi}(1 - a\pi)$: S rises on $[0, 1/a]$, falls after, $S(0) = 0$, $S \to 0$ as $\pi \to \infty$. Maximum at $\pi = 1/a = 40\%$: $d_{max} = \kappa/(ae) = 4.41\%$ of GDP.

Step — two roots. For $0 < d < d_{max}$, continuity and strict monotonicity on each branch give exactly one root on each side of $1/a$ (intermediate value theorem): $\pi_L = 14.3\%$, $\pi_H = 86.1\%$.

Ref: Romer (2012), §11.9, eq. (11.68), pp. 567–572; Kurlat (2020), §11.3, eq. (11.3.1), p. 216.

Grading rubric.

- Steady-state condition: 0.5 pt.
- Laffer maximum and d_{max}: 1 pt.
- Exactly two roots, with the monotonicity argument: 1 pt. Roots found numerically without the argument: 0.5 pt.
- Values: 0.5 pt.

b) (4 points) Expectations adapt: $\dot{\pi}^e = \gamma(\pi - \pi^e)$. Derive $\dot{\pi}^e = \frac{\gamma}{1-a\gamma} h(\pi^e)$ with $h(\pi) \equiv \frac{d}{\kappa} e^{a\pi} - \pi$. Prove that if $a\gamma < 1$ the low-inflation steady state is stable and the high one unstable, and that this reverses if $a\gamma > 1$. What happens under perfect foresight? Evaluate with $\gamma = 0.3$.

Step — derivation. $m = \kappa e^{-a\pi^e} \Rightarrow \dot{m}/m = -a\dot{\pi}^e$. Money market: $\dot{m}/m = d/m - \pi$. Adaptive expectations: $\pi = \pi^e + \dot{\pi}^e/\gamma$. Hence $-a\dot{\pi}^e = d/m - \pi^e - \dot{\pi}^e/\gamma$, i.e. $\dot{\pi}^e(\frac{1}{\gamma} - a) = \frac{d}{\kappa} e^{a\pi^e} - \pi^e$.

Step — stability. At a steady state $\frac{d}{\kappa} e^{a\pi_s} = \pi_s$, so $h'(\pi_s) = a\frac{d}{\kappa} e^{a\pi_s} - 1 = a\pi_s - 1$: negative at π_L ($a\pi_L < 1$), positive at π_H. The local eigenvalue is $\frac{\gamma}{1-a\gamma} h'(\pi_s)$: with $a\gamma < 1$, π_L is stable and π_H unstable; with $a\gamma > 1$ the signs flip.

Step — numbers. $a\gamma = 0.75$, $\frac{\gamma}{1-a\gamma} = 1.2$; $h'(\pi_L) = -0.643$, $h'(\pi_H) = 1.153$: eigenvalues -0.771 and $+1.384$.

Step — perfect foresight. $\pi = \pi^e$ gives $\dot{\pi} = -h(\pi)/a$: π_L unstable, π_H stable. Since the price level can jump, every path starting with $\pi > \pi_L$ converges to π_H (between the roots $h < 0$, above π_H $h > 0$): a continuum of equilibria, and the low-inflation steady state is reached only by starting exactly there.

Ref: Romer (2012), §11.9, eq. (11.75) and footnote 34, and problem 11.22, p. 583.

Grading rubric.

- Derivation of the law of motion: 1.5 pts.
- Stability with $h'(\pi_s) = a\pi_s - 1$: 1.5 pts.
- Perfect foresight and indeterminacy: 1 pt.

c) (4 points) Let $d > d_{max}$. Prove that $h(\pi) > 0$ for every π, that π^e then rises without bound when $a\gamma < 1$, and that it reaches infinity in finite time. What goes wrong at $a\gamma = 1$?

Step — minimum of h. $h'(\pi) = a\frac{d}{\kappa} e^{a\pi} - 1 = 0$ at $\pi_m = \frac{1}{a} \ln \frac{\kappa}{ad}$, a minimum ($h'' > 0$): $h(\pi_m) = \frac{1}{a}[1 - \ln \frac{\kappa}{ad}]$, positive iff $\ln \frac{\kappa}{ad} < 1 \iff d > \frac{\kappa}{ae} = d_{max}$.

Step — no bound. With $a\gamma < 1$, $\dot{\pi}^e = \frac{\gamma}{1-a\gamma} h(\pi^e) \geq \frac{\gamma}{1-a\gamma} h(\pi_m) > 0$: π^e rises at least linearly.

Step — finite time. The time to go from π_0 to infinity is $T = \frac{1-a\gamma}{\gamma}\int_{\pi_0}^{\infty}\frac{d\pi}{h(\pi)}$. For large π, $h(\pi) \geq \frac{d}{2\kappa}e^{a\pi}$, so the integrand is bounded by $\frac{2\kappa}{d}e^{-a\pi}$, which is integrable: $T < \infty$.

Step — $a\gamma = 1$. The coefficient $\gamma/(1-a\gamma)$ is undefined: expectations adjust so fast that the money-market equation no longer determines $\dot{\pi}^e$; the model has no well-defined dynamics at that knife edge.

Ref: Romer (2012), §11.9, pp. 572–576, and problem 11.22.

Grading rubric.

- Minimum of h and the equivalence with $d > d_{max}$: 1.5 pts.
- Unbounded growth: 1 pt.
- Finite-time argument (an integrable bound): 1 pt. "It explodes" without an argument: 0 for this line.
- Knife edge: 0.5 pt.

d) (4 points) The deficit rises from 3% to 5% of GDP while the economy sits at π_L, with $\gamma = 0.3$. The time for expected inflation to reach 100% a year is 6.58 years and to diverge, 6.82 years (numerical integration of (c)). Explain where the time goes, using the minimum of h. What does indexation that raises γ above $1/a$ do, and which policies end the process?

Step — bottleneck. $\pi_m = \frac{1}{2.5}\ln\frac{0.3}{2.5\cdot 0.05} = 35.0\%$ and $h(\pi_m) = 0.0498$: $\dot{\pi}^e$ is smallest near 35%, where the two steady states used to be. Most of the 6.8 years is spent crawling past that point; the final rise from 100% to infinity takes only 0.24 years.

Step — indexation. $\gamma > 1/a = 0.4$ a year flips stability: even with $d < d_{max}$ the low-inflation steady state becomes unstable, so inflation drifts to the high steady state or beyond (inertial inflation).

Step — policy. Only bringing d below d_{max} restores steady states; a credible stabilization must cut the deficit and reset expectations (de-indexation), as the 1994 Real Plan combined fiscal adjustment with the URV. A higher money growth rate cannot raise revenue past d_{max}.

Economic intuition. Money printing is a tax with a shrinking base. Past the Laffer peak, no inflation rate pays the bill, and expectations turn a fiscal gap into an explosion.

Ref: Romer (2012), §11.9; Kurlat (2020), §11.3 and Exercise 11.6.

Grading rubric.

- Bottleneck from the minimum of h: 1.5 pts.
- Indexation and the stability flip: 1.5 pts.
- Policy: 1 pt. "Print more money" as a remedy: 0 for this line.